

International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B102 Foundations of Computer and Digital Forensics

LAB 4

USB Image Acquisition and Hash Verification

Student Practical Guide | Windows FTK Imager | Linux DD | Evidence Integrity

Course Code	CIP-B102
Course Title	Foundations of Computer and Digital Forensics
Lab Number	Lab 4
Recommended Duration	3-4 hours practical + report writing
Submission Format	One PDF report with screenshots, image details, hashes, and reflection
Prepared For	ICDFA Trainees

Important pre-lab instruction

Trainees are advised to visit the course materials, read the USB Image Acquisition slide carefully, and understand the concepts before proceeding to this practical lab. Do not start the lab until you understand what a disk image is, why hashing is required, and the difference between imaging a USB and merely copying files.

1. Lab Overview

Scenario: You are acting as a junior digital forensic trainee. A USB flash drive has been submitted for examination. Your duty is to document the evidence, acquire a forensic image of the USB, calculate hashes, verify integrity, and prepare a repeatable report.

Learning Outcomes

- Explain what a disk image is and why it is used in digital forensics.
- Prepare a test USB evidence drive using a controlled file.
- Acquire a USB image using either FTK Imager on Windows or the dd command on Linux.
- Calculate and compare hash values for integrity verification.
- Document the acquisition process with clear screenshots and evidence notes.
- Identify risks when using low-level imaging tools and avoid imaging the wrong disk.

Source Material Covered

- What a disk image is and why forensic acquisition is different from normal file copying.
- USB acquisition in Windows using FTK Imager.
- USB acquisition in Linux using dd and dcfldd.
- Optional network acquisition concept using netcat between Kali and Windows.
- Hash verification of the original media and acquired image.

2. Required Tools and Environment

Item	Requirement	Purpose
USB Flash Drive	Any spare/test USB drive	Evidence media for imaging
Windows PC	Local host PC with administrator access	FTK Imager acquisition option
FTK Imager	FTK Imager 4.5 or newer	Create forensic image and calculate hashes
Linux/Kali VM or Host	Kali Linux, Ubuntu, or customized forensic VM	dd/dcfldd acquisition option
VirtualBox USB Support	Guest Additions / USB passthrough configured if using VM	Allow Linux VM to detect the USB
Hash Utilities	md5sum, sha256sum, certutil, or FTK built-in hashing	Verify integrity
Report Tool	Microsoft Word / Google Docs / LibreOffice	Prepare final lab report

Useful Links from the Material

- FTK Imager download page: <https://accessdata.com/product-download/ftk-imager-version-4-5>
- Windows dd utility: <http://www.chrysocome.net/downloads/dd-0.5.zip>
- Customized digital forensics lab VM/resources: <https://github.com/frankwxu/digital-forensics-lab>
- VirtualBox USB passthrough reading 1: <https://www.linuxbabe.com/virtualbox/access-usb-from-virtualbox-guest-os>
- VirtualBox USB passthrough reading 2: <https://www.net-usb.com/virtual-usb/virtualbox-usb-passthrough/virtualbox-usb-linux/>

Safety warning

Use only a personal test USB drive. Do not image another person's device or an organizational device without written authorization. Do not run dd against your internal system disk. A wrong if= or of= value can destroy data.

3. Evidence Handling Rules

- Treat the USB as evidence once the test file has been created.
- Record who handled the USB, the date/time, and the purpose of handling.
- Avoid modifying the USB after acquisition begins.
- Where possible, use read-only access or a write blocker. If no write blocker is available, clearly state this limitation in the report.
- All screenshots must show enough context: command used, disk selected, hash result, file path, or tool window.
- All findings must be repeatable by another trainee using the same steps.

4. Lab Tasks

Task 1 - Prepare the USB Evidence Media

1. Insert your test USB flash drive into your PC.
2. Create a folder on the USB named CIP-B102-Lab4-Evidence.
3. Inside the folder, create a text file named `firstname_lastname.txt`. Example: `aminu_idris.txt`.
4. Add the following content into the text file, replacing the placeholders with your own details:

```
Evidence File for CIP-B102 Lab 4
Student Name: <Your Full Name>
Registration Number: <Your ID>
Date Created: <Date>
Statement: This USB was prepared for a controlled forensic acquisition lab.
```

5. Optional: add one image file and one PDF file to the USB so that the image contains more than one file type.
6. Safely eject the USB, then reconnect it for acquisition.

Required Screenshot Set A

- USB drive visible in File Explorer or Linux file manager.
- The created folder and `firstname_lastname.txt` file.
- The content of `firstname_lastname.txt`.

Task 2 - Complete the Evidence Acquisition Form

Before imaging, complete the following table in your final report.

Field	Student Entry
Case/Lab ID	
Student Name	
USB Brand/Description	
USB Capacity	
Date and Time Received/Prepared	
Acquisition Method Used	
Tool Used and Version	
Image File Name	
Image Storage Location	
Hash Algorithm Used	
Examiner Signature	

Task 3 - Choose Your Acquisition Method

Students must complete at least one of the following acquisition options. Option A is recommended for Windows users. Option B is recommended for Linux/Kali users. Option C is an optional challenge for students who have two lab machines on the same network.

Option A - Windows Acquisition Using FTK Imager

7. Download and install FTK Imager from the FTK Imager link provided in the course material.
8. Run FTK Imager as Administrator.
9. Go to File > Create Disk Image.
10. Select Physical Drive if you want the whole USB, or Logical Drive if instructed to image only the visible volume.
11. Carefully select your USB drive. Confirm the size and drive label before continuing.
12. Choose the image type. Raw/dd is recommended for this beginner lab because it is simple and works with many forensic tools.
13. Enter evidence item information such as case number, evidence number, examiner, and description.
14. Choose an output folder on your computer. Do not save the image back onto the same USB.
15. Name the image using this format: CIPB102_Lab4_Firstname_Lastname_USB.dd
16. Enable verification after image creation if available.
17. Start the acquisition and wait until it completes.
18. Save or screenshot the hash values generated by FTK Imager.

Required Screenshot Set B - FTK Method

- FTK Imager Create Disk Image screen.
- Selected USB drive details showing size/label.
- Image type and destination path.
- Acquisition progress or completion screen.
- Final hash/verification result.

Option B - Linux/Kali Acquisition Using dd

Critical warning for dd

In dd, if= means input file/source and of= means output file/destination. Selecting the wrong device can overwrite important data. Check the device name carefully before running the imaging command.

19. Open a terminal in Linux/Kali.
20. List disks before inserting the USB:

```
lsblk
sudo fdisk -l
```

21. Insert the USB and list disks again to identify the new device. Example: /dev/sdb or /dev/sdc.

```
lsblk
sudo fdisk -l
```

22. Create a folder to store your acquired image:

```
mkdir -p ~/CIP-B102-Lab4
```

23. If the USB auto-mounted, unmount its partition before imaging. Replace /dev/sdX1 with the correct partition name:

```
sudo umount /dev/sdX1
```

24. Acquire the USB image. Replace /dev/sdX with the whole USB device name, not your internal disk:

```
sudo dd if=/dev/sdX of=~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd bs=4M status=progress
conv=noerror,sync
```

25. Flush cached writes and confirm the image file exists:

```
sync
ls -lh ~/CIP-B102-Lab4/
```

26. Calculate hash values of the original USB device and the acquired image:

```
sudo md5sum /dev/sdX
md5sum ~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd

sudo sha256sum /dev/sdX
sha256sum ~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd
```

Optional: Using dcfldd

dcfldd can display progress and calculate hashes during acquisition on some systems. Install it if available, then run a controlled command similar to the following:

```
sudo apt update
sudo apt install dcfldd -y

sudo dcfldd if=/dev/sdX of=~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd hash=md5,sha256
hashlog=~/CIP-B102-Lab4/hashlog.txt
```

Required Screenshot Set C - Linux/dd Method

- lsblk or fdisk output before and after USB insertion.
- Unmount command, if applicable.
- dd or dcfldd acquisition command running or completed.
- Image file shown with ls -lh.
- MD5 and SHA256 hash results for original USB and acquired image.

Option C - Optional Challenge: Network Acquisition Concept

Instructor approval required

This task must only be done in an isolated classroom/lab network using a test USB device. Do not send disk images across public networks or networks you do not control.

Goal: acquire a USB image from Linux/Kali and transfer it to a Windows or Linux receiving machine over the lab network, then verify the hash at both ends.

27. On the receiving machine, start a listener and save incoming data to a file. Example on Linux receiver:

```
nc -lp 8080 > CIPB102_Lab4_network_USB.dd
```

28. On the acquisition machine, send the USB image through netcat. Replace /dev/sdX and receiver IP address:

```
sudo dd if=/dev/sdX bs=4M status=progress conv=noerror,sync | nc <receiver-ip-address> 8080
```

29. Calculate hashes on both machines and compare them:

```
md5sum CIPB102_Lab4_network_USB.dd
sha256sum CIPB102_Lab4_network_USB.dd
```

- In your reflection, explain why network acquisition can be slower and why hash verification is important after network transfer.

5. Hash Verification Table

Complete this table in your final report. Hashes must match between the original USB source and the acquired image for integrity verification. If they do not match, explain possible causes and repeat the acquisition if necessary.

Evidence Item	MD5 Hash	SHA256 Hash	Match/Comment
Original USB / source device			
Acquired image file			
Recovered/mounted image check			

6. Validate the Image Contents

After acquisition, students should show that the image contains the file they placed on the USB. Use any one safe method below.

Method 1 - FTK Imager

30. Open FTK Imager.
31. Add Evidence Item > Image File.
32. Select your acquired .dd image.
33. Browse the image and locate firstname_lastname.txt.
34. Preview the file content and capture a screenshot.

Method 2 - Linux Mount Read-Only

Read-only analysis

Mount the image read-only. Do not modify the acquired image during validation.

```
mkdir -p ~/CIP-B102-Lab4/mount
sudo mount -o ro,loop ~/CIP-B102-Lab4/CIPB102_Lab4_Firstname_Lastname_USB.dd ~/CIP-B102-Lab4/mount
ls -la ~/CIP-B102-Lab4/mount
cat ~/CIP-B102-Lab4/mount/CIP-B102-Lab4-Evidence/firstname_lastname.txt
sudo umount ~/CIP-B102-Lab4/mount
```

If the image contains a partition table and direct loop mounting fails, students should record the error and use FTK Imager, Autopsy, or instructor guidance to mount the correct partition offset.

7. Final Report Structure

- Cover page: course code, lab number, student name, registration number, date.
- Pre-lab reading confirmation: a short paragraph confirming that you reviewed the course material before starting.
- Evidence acquisition form: complete all required fields.
- Tool and environment description: OS, tool version, USB description, acquisition method.
- Step-by-step procedure: write what you did, not only screenshots.
- Screenshots: include all required screenshot sets for your chosen method.
- Hash verification table: include MD5 and SHA256 values.
- Image validation: prove the image contains firstname_lastname.txt.
- Reflection: explain lessons learned, mistakes avoided, and why hashing matters.
- Conclusion: state whether acquisition was successful and whether integrity was verified.

8. Submission Checklist

No.	Requirement
1	I read and understood the USB Image Acquisition material before beginning the lab.
2	I created firstname_lastname.txt on a test USB drive.
3	I documented the USB description, capacity, and acquisition method.
4	I acquired a forensic image using FTK Imager, dd, dcfldd, or an instructor-approved method.
5	I calculated at least MD5 and SHA256 hash values.
6	I compared the source and image hashes.
7	I proved that the acquired image contains my evidence file.
8	I included clear screenshots for each major step.
9	I submitted one final PDF report by the deadline.

9. Marking Rubric - 100 Marks

Assessment Area	Marks	What the Instructor Will Check
Pre-lab preparation and understanding	10	Student confirms reading and explains disk image concept correctly.
Evidence preparation and documentation	15	USB evidence file created, evidence form completed, handling notes are clear.
Correct acquisition procedure	25	Student selects correct device and performs imaging using FTK/dd/dcfldd safely.
Hashing and integrity verification	20	MD5/SHA256 calculated, compared, and interpreted correctly.
Image content validation	10	Student proves the image contains the expected evidence file.
Screenshots and repeatability	10	Screenshots are complete, readable, and aligned with the written steps.
Final report quality	10	Report is organized, professional, and includes reflection and conclusion.

10. Common Mistakes to Avoid

- Imaging the wrong disk because the USB was not identified carefully.
- Saving the image file onto the same USB being acquired.
- Forgetting to calculate hashes before and after acquisition.
- Submitting only screenshots without explaining what was done.
- Using a real/private USB containing sensitive personal data.
- Running destructive commands without instructor approval.
- Failing to mention limitations such as no write blocker being available.

11. Student Declaration

I confirm that this lab was performed only on a test USB device that I am authorized to use. I understand that forensic acquisition tools must not be used on any device without permission.

Student Name	
Signature	
Date	